KU LEUVEN

# Lab seminar

## Doing research with AI agents

Rigour made affordable: a knowledge tree of inquiries

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FLANDERS MAKE
DRIVING INNOVATION IN MANUFACTURING

RELIABLE & ROBUST DESIGN

# What happens when an agent does research unchecked?

### ! It cites what it never read

Plausible references and values that do not exist.

*Walters & Wilder 2023*

### ! It tries forty variants, reports one

The garden of forking paths, now nearly free.

*Gelman & Loken 2014*

### ! It agrees with you

Models tend to tell users what they want to hear.

*Sharma et al. 2024*

**Research has long known the cure: preregistration, verification, published negatives, review. It could rarely afford all of it.**

# Agents make rigour affordable: twelve rules, four groups

## 1 Planning

Fix the protocol before the runs: explore or confirm.

**fixes:** *testing what the data suggested*

## 2 Evidence

Every number traces to a run, a derivation or a page.

**fixes:** *numbers nobody can rebuild*

## 3 Claims

Publish the negatives and the denominator; cite the page you checked.

**fixes:** *forty variants, one reported; invented citations*

## 4 The tree

The human confirms, after a blind review.

**fixes:** *the agent agreeing with you*

**12 rules, 5 of them fixed. The agent shows them to you at the start, and asks before you set one aside.**

*Grounded in the reproducibility literature: Ioannidis 2005; Munafò et al. 2017; full map in docs/literature.md.*

# Anatomy of an agent: own the top layer

**Your framework**
*the memory*

Plain Markdown in your folder: AGENTS.md · .agents/skills/ · docs/wiki/ · inquiry.yaml

**Harness**
*the hands*

Reads and writes files, runs code, loops until done: Claude Code · Codex · Gemini CLI · OpenCode

**LLM**
*the brain*

Predicts the next token; knows nothing of your project: Claude · GPT · Gemini · Qwen

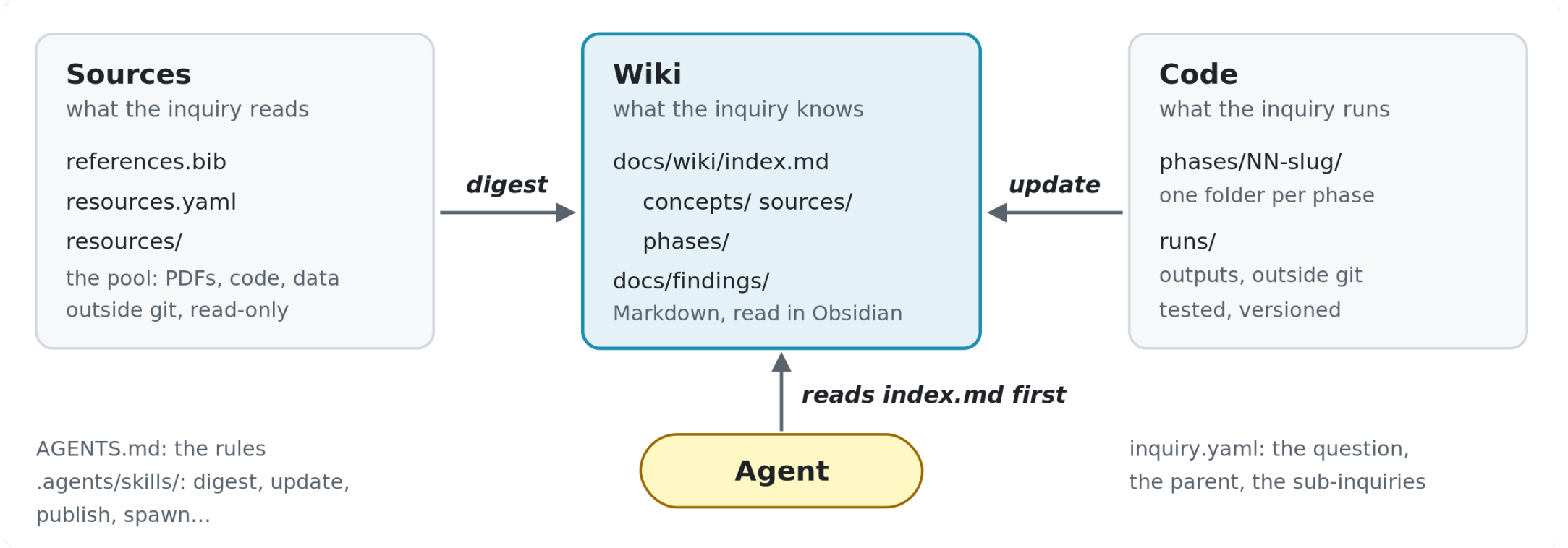
## Own the top layer

Models and harnesses change every few months, and so do the leaders.

Keep what you build in plain Markdown in your folder: switch model or harness, keep your knowledge.

AGENTS.md is read by most harnesses; Claude Code reads it through CLAUDE.md.

# One folder, one question: sources, wiki, code




**Sources are digested into the wiki, code changes update it, and the agent reads index.md first and cites the notes it used.**

# The wiki has two readers

## You, in Obsidian

Browse from index.md, follow the links

The graph: how sources and concepts connect

“What did we take from X?” in seconds

## The agent, as context

Reads index.md before answering

Lands on the 2–3 right notes, not 40 PDFs

Cites the notes, so you can check

*illustrative*

```
# Kooijman et al. (2011): a bicycle stable
#   without gyroscopic or caster effects

[@kooijman2011bicycle]. A purpose-built bicycle
with neither effect still balances itself.

## What we take
- **Neither effect is necessary.**
  → [gyroscopic](../concepts/gyroscopic.md)

## What we don't take
Their prototype's mass layout, for now.
```

*One note per source or concept. A digest is a selection, not a summary: what the inquiry uses or rejects, with a link to where.*

# When a question is too broad, it grows a tree

```
bicycle/              why does a moving bicycle
│                     stay upright?
└── inquiries/
    ├── benchmark-model/
    │   └── inquiries/speed-sweep/
    ├── gyroscopic/
    └── trail/
```

**Zoom in**

“Spawn a sub-inquiry for the gyroscopic effect”: a child repository with its own question, under its parent.

**Zoom out**

Adopt an existing repository under a broader question; merging two is a new root and two adopts.

**Knowledge flows**

Down by link, up by findings, one level at a time; never sideways.

**Every index routes to its children, the open questions and what is out of scope: asked a broad question, the agent descends one index at a time.**

# What it looks like

**docs/wiki/index.md, at the root**

```
## Sub-inquiries
- [gyroscopic](…/index.md): how much does
  wheel spin contribute? Active.
  Dive here for: wheel spin, precession.
- [trail](…/index.md): …

### Open
- How much does the rider's steering
  contribute? (no inquiry yet)
```

**docs/findings/…, at the root**

*illustrative*

```
status: hint      # its children are hints
tried: 3          # explanations tested
evidence:
  - kind: sub-inquiry
    finding: …/gyroscopic/blob/8be41d0/…
    status: hint

# Neither wheel spin nor trail alone
# keeps the benchmark bicycle upright
Scope: the model only; not yet validated.
```

**rkt check verifies what a machine can tell: links, routes, the denominator, no parent stronger than its children.**

# Limits

### ! Errors get rarer, not impossible

The rules and the checks make invention harder.

Check what matters against the source.

### ! Confidentiality

Files and prompts go to the model provider.

Keep confidential material out of the inquiry; deny it in your harness's settings.

### ! Cost and lock-in

Subscriptions, quotas and leaders change.

Plain Markdown keeps your work portable.

**You remain the author: of the questions, of the code you merge, and of every claim you sign.**

# The tools will change. Who decides should not.

Models and harnesses improve every few months: this kit may be obsolete within a year.

What stays human is choosing the questions, where to point the tools, and judging what they return.

**Rule 12 builds that in:** you approve every protocol and every confirmed finding. Staying in the loop is part of the method, not a wish.

## Use the agent to learn

Ask it to explain, then explain it back

Derive by hand, then let it check you

Read the digest, then the paper

Let a grilling skill question your plan

The wiki is written for you too: read it

**A result nobody understands, remembers or communicates is wasted computation.**

# Start tomorrow: point your agent at the repo

```
Install research-knowledge-tree from github.com/bonneted/research-knowledge-tree,
make this folder an inquiry, and start it.
My question is: …
```

## Or by hand

```
uv tool install \
  git+https://github.com/bonneted/research-knowledge-tree@v2.0.0
```

`rkt init` then: “start the inquiry”

## You need

An agent: Claude Code, Codex, Gemini CLI…

Obsidian, to read the wiki

Zotero: recommended, not required

**Start small: one inquiry, five papers, one question.**

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